

SALIM SADMAN

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Mechanical engineer focused on robotic hardware, actuator design and mechatronic system integration. Designed and experimentally characterized a custom cycloidal quasi-direct-drive (QDD) actuator and built the 12-DoF quadruped platform it drives for a B.Sc. thesis on muddy-terrain locomotion. Co-led a university rover team through the Anatolian Rover Challenge 2026 finals, where it took the competition's highest design score.

Research interests: Robot design and fabrication · Actuator design and motor control · Legged locomotion on deformable terrain · Field, space and marine robotics · Mechatronic system integration

EDUCATION

B.Sc. in Mechanical Engineering

Mar 2022 – Sep 2026

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

CGPA: 3.56 / 4.00

Thesis: *Design and Development of a Quadruped Robot for Optimizing Muddy Terrain Locomotion*

Supervisor: Prof. Md. Emdadul Haque, Department of Mechanical Engineering, RUET

HSC (Science), Dhaka College, 2020, and **SSC (Science)**, B.L. Govt. High School, Sirajganj, 2017 — GPA 5.00 / 5.00 in both

RESEARCH & ENGINEERING EXPERIENCE

Quadruped Robot with Custom QDD Actuators for Muddy-Terrain Locomotion

Ongoing

B.Sc. thesis research, RUET: actuator design, mechanical design, fabrication and bench testing

- Designed a compact **quasi-direct-drive joint actuator**: BLDC motor, cycloidal reducer, encoder feedback and field-oriented control (FOC), for the torque density and backdrivability that legged locomotion needs
- Fabricated and assembled all twelve actuators and the robot's full **12-DoF structure** in-house
- Built a **motor-characterization test bench** (ODrive v3.6 + ESP32) whose Python pipeline time-aligns controller telemetry with output-side measurements to map efficiency
- Designed the platform to carry **IMU and depth-camera sensing** for locomotion experiments on mud

MEER 2.0 Mars Rover, Team Ogrodoot

Oct 2024 – Sep 2026

Co-Lead & Mechanical Sub-Team Lead (Aug 2025 – Sep 2026); Mechanical Design & Fabrication (Oct 2024 – Aug 2025)

- Designed the **4-wheel rocker-bogie chassis**, differential suspension, custom wheel assemblies and **5-DoF robotic arm**
- Designed the actuator mechanisms and a **motor-driven lead-screw gripper** for payload manipulation
- Co-led the team through the **Anatolian Rover Challenge (ARC) 2026 finals** — 13th overall, with the competition's **highest design score** — while heading its mechanical sub-team
- Ran **20+ mentoring sessions** (2025–26) for junior members on mechanical-team practice and CAD design in Fusion 360; closely involved in recruiting new members

Underwater Glider for Aquatic Environmental Monitoring

- Developed an underwater glider with **buoyancy-based depth control**, integrating dissolved-oxygen, ORP, turbidity and TDS sensors for water-quality monitoring

IoT Engineer (Hardware Engineering)

Jan 2024 – Jan 2025

SystemSage Solutions Sdn. Bhd.

- Built the entire hardware side of **AgroSense**, a team smart-farming system **published at GEn-CITY 2024 [P1]**: ESP32 field nodes reading a 7-in-1 soil sensor (NPK, pH, moisture, temperature, conductivity) over LoRa to a gateway that uploads to a cloud dashboard over 4G
- Designed its **PCBs, wiring and firmware**, prototyped several GSM/4G modules, and built the relay control module that switches pumps and lights manually, on range-based rules or on a schedule
- Engineered **solar-powered nodes** for remote deployment: BME280 ambient sensing, LoRa, Wi-Fi and GPRS links

PUBLICATIONS

[P1] M. H. A. Afif, J. Iqbal, **Salim Sadman Borshon**, R. I. Junayed, and S. Ahamed, "An IoT-based smart farming solution for sustainable agriculture: integrating AI, cloud computing, and 4G communication for enhanced productivity," in *Proc. Int. Conf. on Green Energy, Computing and Intelligent Technology (GEn-CITY 2024)*, Malaysia (virtual), Dec. 2024. *IET Conference Proceedings*, vol. 2024, no. 30, pp. 83–88. doi: [10.1049/icp.2025.0238](https://doi.org/10.1049/icp.2025.0238)

TEACHING, MENTORING & OUTREACH

Robotics Instructor

Jan 2025 – Present

Artificial Intelligence Bangladesh (AIBD); full-time since Aug 2026

- Taught **80+ classes with 500+ student engagements** in robotics, electronics and programming, and helped organize **five regional and national robotics events**, initially alongside full-time undergraduate study
- Flagship programs: a 15-day robotics and 3D-printing workshop at Bangladesh Computer Council, Rajshahi; a 4-day robotic-hand workshop at American Corner Rajshahi; a robotic-hand session for IEEE RAS, University of Rajshahi
- Wrote the organization's **course modules** — electronics, robotics, line-following robots, C, Python (including object-oriented programming) and, most recently, CAD and 3D printing — plus the illustrated assembly manuals for its line-follower and smart-dustbin kits

Workshop Instructor and Organizer

2024 – 2025

Robotic Society of RUET

- Organized and taught three workshops (six classes, **200+ student engagements**) on robotics, electronics, microcontrollers and PCB design

LEADERSHIP & SERVICE

Robotic Society of RUET

Nov 2023 – Aug 2026

General Secretary (2025–26) · Assistant Technical Secretary (2024–25) · Assistant Technical Coordinator (2023–24)

- Secured sponsorship for and organized “Ro Te Robot”, a national robotics event; ran three annual recruitment drives

IEEE RUET Student Branch

Feb 2025 – Sep 2026

Sponsorship Coordinator (2025–26) · Content Writer (2025)

- Raised funding that launched several branch programs; ran a 2025 STEM outreach event at B.L. Govt. High School

HONORS & AWARDS

Highest Design Score and Finalist, 13th overall — Anatolian Rover Challenge (ARC) 2026 (Team Ogrodoot)	2026
Semi-finalist , International Rover Design Challenge (IRDC), Space Robotics Society (Team Ogrodoot)	2025
1st Place , Poster Presentation, RUET Innovista	2025
2nd Place , Project Showcase, Astro Science Fair, BSMR Novo Theatre, Rajshahi	2024
1st Place , Line-Following Robot Competition, Tech-Mayhem, Robotic Society of RUET	2023

TECHNICAL SKILLS

Mechanical design: Fusion 360 (including Python API scripting); mechanism design (cycloidal reducers, rocker-bogie suspension, lead screws, grippers); FDM 3D printing in ABS and TPU

Actuation & control: BLDC motors, field-oriented control (FOC), ODrive, encoder feedback, PID control, motor characterization and test-bench design

Embedded & electronics: ESP32, STM32, Arduino; PCB design (EasyEDA); IMU, depth-camera, RS485 and water-quality sensor integration; LoRa, Wi-Fi, GPRS/4G LTE; solar power systems

Programming: C/C++, Python (data acquisition and telemetry pipelines), MATLAB

TRAINING & INDUSTRIAL EXPOSURE

Industrial attachment — Walton Hi-Tech Industries PLC, Dept. of Engineering Services Management, Chandra, Gazipur (2–25 March 2025): studied high-volume manufacturing end to end, from refrigeration and HVAC assembly, PCB fabrication, battery packs and consumer electronics to plastics recycling and the R&D section, with process and design choices discussed directly with the engineers running each line

Basic Programming with Python, 60-hour course (Dec 2024 – Feb 2025): IICT, RUET, under the EDGE project (Bangladesh Computer Council, ICT Division); fundamentals through introductory machine learning

Study tour (2025): 670 MW combined-cycle dual-fuel power plant, organized by IEEE RUET Student Branch